GUY MOSS

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Guy Moss | gmoss13 | guymoss13

Graduating Machine Learning Ph.D. working on Statistical Inference, Generative Modeling, and Geoscience.

Education

- **Ph.D., Machine Learning in Science** — *10/2021 - 11/2025 (expected)*
  University of Tübingen and International Max Planck Research School for Intelligent Systems — Tübingen, Germany
- **Master of Mathematics and Physics (MMathPhys)** — *10/2017 - 07/2021*
  University of Warwick — Warwick, United Kingdom
  - First Class with Honours. Weighted mean mark—86.2%

Experience

- **Graduate Researcher** — *Tübingen, Germany*
  University of Tübingen — 10/2021 - Present
  - Trained deep probabilistic generative models such as normalizing flows and diffusion models to solve Bayesian inference problems.
  - Developed new approach combining neural operators and flow matching to solve inference problems for thousands of parameters.
  - Collaborated with geoscientists to establish a new machine learning approach for inferring the melting rates of ice in Antarctica.
- **Core Developer** — *Tübingen, Germany*
  sbi — 01/2022 - Present
  - Maintained open source code used by hundreds of practitioners (3000 monthly downloads on PyPI).
  - Implemented features for simulation-based inference with diffusion models and deployed them into production via CI/CD workflows.
  - Organized hackathons involving 40+ contributors.
- **Graduate Teaching Assistant** — *Tübingen, Germany*
  University of Tübingen — 10/2021 - Present
  - Taught classes in graduate courses including > 100 students at MSc level.
  - Supervised several MSc thesis projects in probabilistic machine learning and geophysics.
- **Research Intern** — *Warwick, United Kingdom*
  University of Warwick — 07/2019 - 09/2019
  - Developed data analysis pipeline for timeseries astronomical data using Astropy.
  - Applied Markov Chain Monte Carlo (MCMC) to infer magnetic activity in sun-like stars.

Skills

- **Machine Learning:** Bayesian Inference, Diffusion Models, Neural Operators, Variational Inference
- **Software:** Python, PyTorch, git, Hydra, MATLAB , JAX
- **Scientific Computing:** Finite Difference and Finite Element solvers, (Neural) Differential Equations
- **Languages:** English (Native), Hebrew (Native), Russian (Proficient), German (Conversational)
- **Icebreakers:** International level chess player, Former student radio show host

Awards

- **Pettifer Prize for best first year student** — *2018*
  University of Warwick Physics Department
- **Excellence Award for best graduating student** — *2021*
  University of Warwick Physics Department

Selected Publications

C=Conference, J=Journal

[C] **Guy Moss & Julius Vetter** et al. (2024), **Sourcerer: Sample-Based Maximum Entropy Source Distribution Estimation**. In *NeurIPS*.

[J] **Guy Moss** et al. (2025), **Simulation-Based Inference of Surface Accumulation and Basal Melt Rates of an Antarctic Ice Shelf from Isochronal Layers**. In *Journal of Glaciology*.

[C] **Guy Moss** et al. (2025), **FNOPE: Simulation-based inference on function spaces with Fourier Neural Operators.** In *NeurIPS*.